[*] Untitled1 | Untitled2.cpp

```c
#include<stdio.h>
int main()
{
    int products[5][2];
    int i,j,total,grandTotal=0;

    for(i=0; i<=4; i++)
    {
        printf("enter quantity of product %d ",i+1);
        scanf("%d",&products[i][0]);

        printf("enter unit price of product %d ",i+1);
        scanf("%d",&products[i][1]);
    }

    printf("\nproducts quantity price total\n");
    public int __cdecl printf (const char * __restrict__ _Format, ...)
    for(int i=0;i<5;i++)
    {
        total=0;
        for(j=0;j<2;j++)
        {
          if(j==0)
            total=products[i][j]*products[i][j+1];
        }
        printf("%d\t%d\t%d\t%d\n",i+1,products[i][0],products[i][1],total);
        grandTotal=grandTotal+total;
    }

    printf("grandTotal=%d",grandTotal);
}
```

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```
enter quantity of product 1 2
enter unit price of product 1 20
enter quantity of product 2 3
enter unit price of product 2 20
enter quantity of product 3 4
enter unit price of product 3 35
enter quantity of product 4 3
enter unit price of product 4 46
enter quantity of product 5 5
enter unit price of product 5 60

products quantity price total
1       2       20      40
2       3       20      60
3       4       35      140
4       3       46      138
5       5       60      300
grandTotal=678
--------------------------------
Process exited after 22.04 seconds with return val
0
Press any key to continue . . .
```

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*] Untitled1 × Untitled2.cpp ×

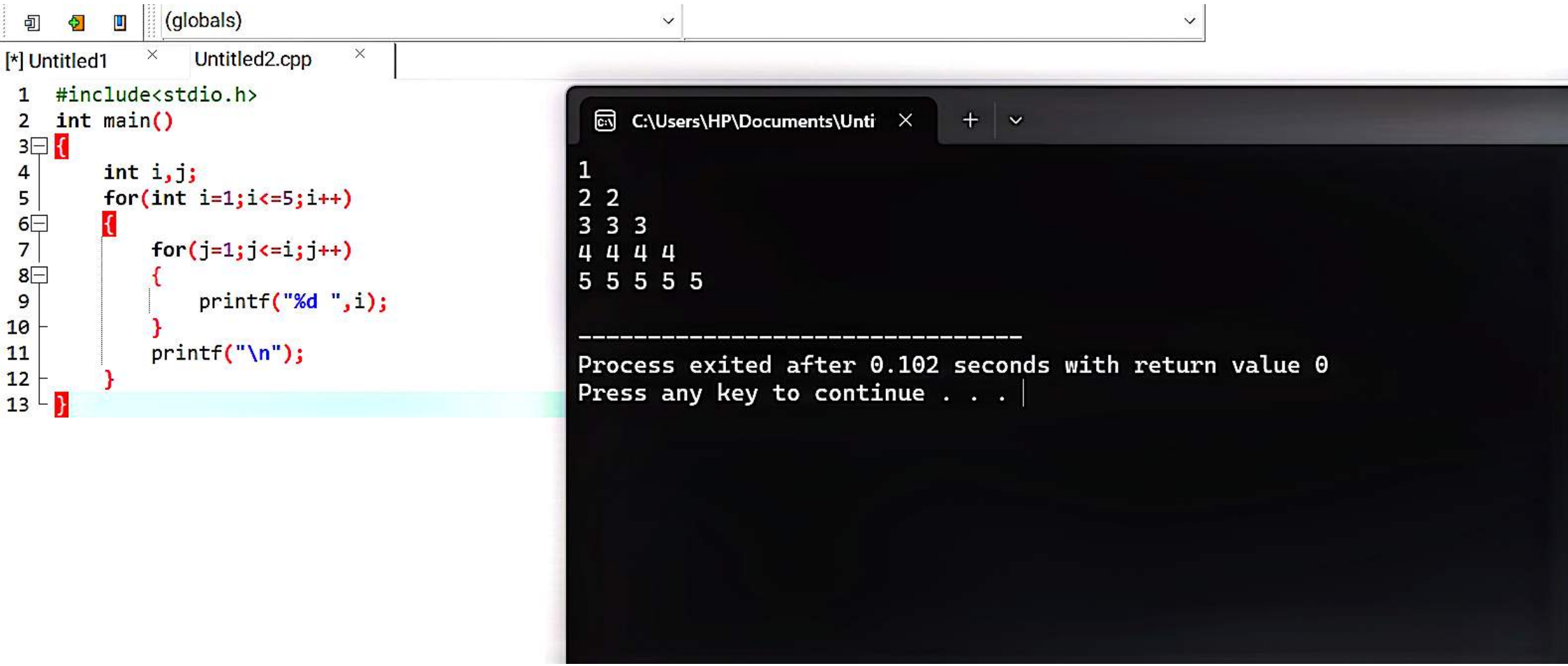
```
1  #include<stdio.h>
2  int main()
3  {
4      int i,j;
5      for(i=1;i<=5;i++)
6      {
7          for(j=1;j<=i;j++)
8          {
9              if(j%2==0)
10             {
11                 printf("\t%d",j*j);
12             }
13             else{
14                 printf("\t%d",j*j*j);
15             }
16
17             }
18                 printf("\n");
19         }
20
21 }
22
```

C:\Users\HP\Documents\Unti × +

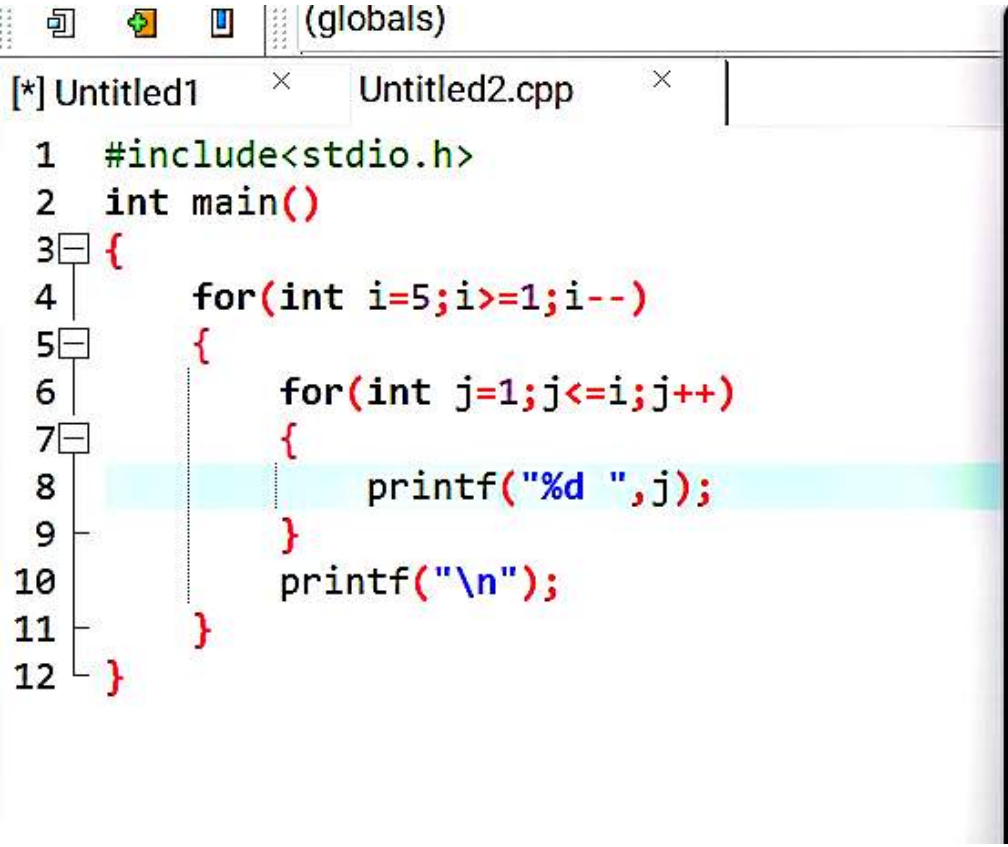
```
        1
        1       4
        1       4       27
        1       4       27      16
        1       4       27      16      125

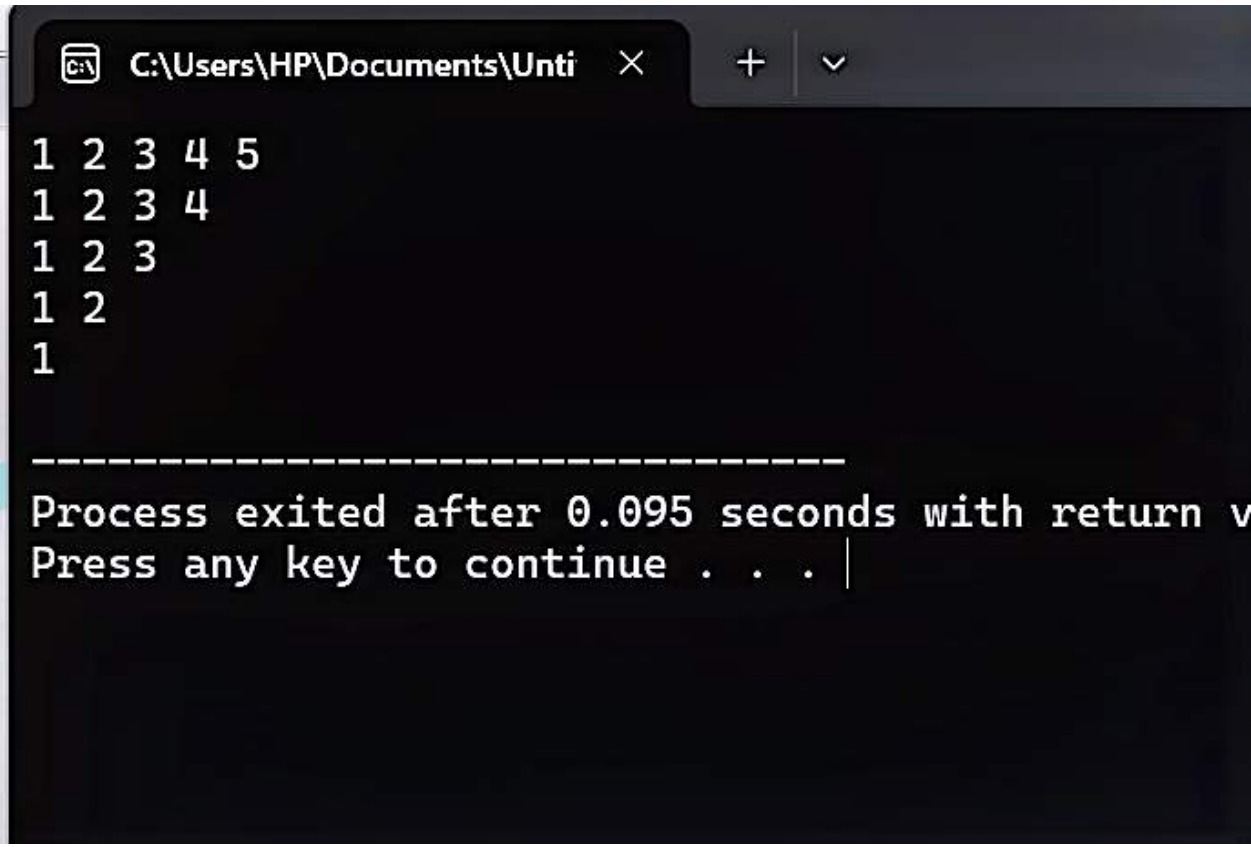
--------------------------------
Process exited after 0.1049 seconds with return value 0
Press any key to continue . . .
```

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(globals)
[*] Untitled1
Untitled2.cpp
#include<stdio.h>
int main()
{
int i,j;
for(int i=1;i<=5;i++)
{
for(j=1;j<=i;j++)
{
printf("%d ",i);
}
printf("\n");
}
}
C:\Users\HP\Documents\Unti
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
Process exited after 0.102 seconds with return value 0
Press any key to continue . . .

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(globals)
[*] Untitled1
Untitled2.cpp
#include<stdio.h>
int main()
{
for(int i=5;i>=1;i--)
{
for(int j=1;j<=i;j++)
{
printf("%d ",j);
}
printf("\n");
}
}


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1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
Process exited after 0.095 seconds with return v
Press any key to continue . . .

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```c

#include <stdio.h>

int main()
{
    int a[3][3];
    int i, j;
    int isNull = 1, diagonal = 1;
    int upper = 1, lower = 1, identity = 1;

    printf("Enter 9 elements of the matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nMatrix is:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            printf("%d ", a[i][j]);
        }
        printf("\n");
    }

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            if (a[i][j] != 0)
                isNull = 0;

            if (i != j && a[i][j] != 0)
```

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```
30      }
31
32      for (i = 0; i < 3; i++)
33      {
34          for (j = 0; j < 3; j++)
35          {
36              if (a[i][j] != 0)
37                  isNull = 0;
38
39              if (i != j && a[i][j] != 0)
40                  diagonal = 0;
41
42              if (i > j && a[i][j] != 0)
43                  upper = 0;
44
45              if (i < j && a[i][j] != 0)
46                  lower = 0;
47
48              if (i == j && a[i][j] != 1)
49                  identity = 0;
50
51              if (i != j && a[i][j] != 0)
52                  identity = 0;
53          }
54      }
55
56      printf("\nMatrix Types:\n");
57      printf("Square Matrix\n");
58
59      if (isNull == 1)
60          printf("Null Matrix\n");
61
62      if (diagonal == 1)
63          printf("Diagonal Matrix\n");
```

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```c
printf("\nMatrix Types:\n");
printf("Square Matrix\n");

if (isNull == 1)
    printf("Null Matrix\n");

if (diagonal == 1)
    printf("Diagonal Matrix\n");

if (upper == 1)
    printf("Upper Triangular Matrix\n");

if (lower == 1)
    printf("Lower Triangular Matrix\n");

if (identity == 1)
    printf("Identity Matrix\n");

printf("\nDiagonal Elements: ");
for (i = 0; i < 3; i++)
{
    printf("%d ", a[i][i]);
}

printf("\nUpper Triangular Elements: ");
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        if (i <= j)
            printf("%d ", a[i][j]);
    }
}

printf("\nLower Triangular Elements: ");
```

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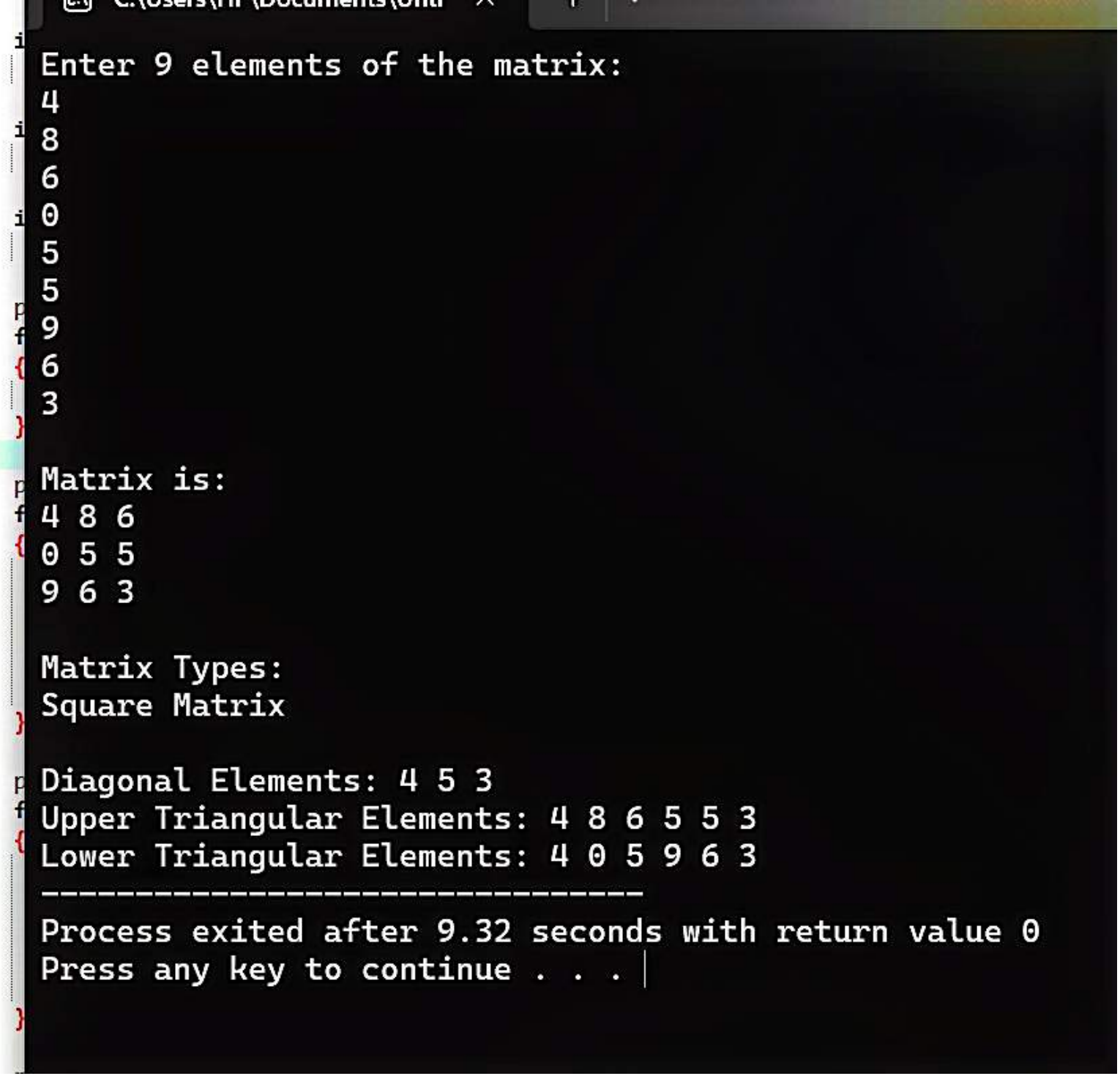
```
78      }
79
80      printf("\nUpper Triangular Elements: ");
81      for (i = 0; i < 3; i++)
82      {
83          for (j = 0; j < 3; j++)
84          {
85              if (i <= j)
86                  printf("%d ", a[i][j]);
87          }
88      }
89
90      printf("\nLower Triangular Elements: ");
91      for (i = 0; i < 3; i++)
92      {
93          for (j = 0; j < 3; j++)
94          {
95              if (i >= j)
96                  printf("%d ", a[i][j]);
97          }
98      }
99
100     return 0;
101 }
102
```







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Enter 9 elements of the matrix:
4
8
6
0
5
5
9
6
3
Matrix is:
4 8 6
0 5 5
9 6 3
Matrix Types:
Square Matrix
Diagonal Elements: 4 5 3
Upper Triangular Elements: 4 8 6 5 5 3
Lower Triangular Elements: 4 0 5 9 6 3

Process exited after 9.32 seconds with return value 0
Press any key to continue . . .

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